



Innovative music education: An empirical assessment of ChatGPT-4's impact on student learning experiences

Wang Zhou¹ · YeaJin Kim¹

Received: 3 October 2023 / Accepted: 10 April 2024 / Published online: 24 April 2024
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2024

Abstract

The application of artificial intelligence technologies, such as ChatGPT-4, offers a substantial chance to improve the learning process. The promising prospect of utilizing artificial intelligence and ChatGPT-4 to enhance educational interactions by offering tailored and engaging experiences is highly enticing. The promise of ChatGPT-4 resides in customizing teaching materials to correspond with the distinct learning objectives and musical preferences of individual students. Nevertheless, only a restricted amount of research has examined the effectiveness of ChatGPT-4 in improving students' learning outcomes. The present study aims to investigate the influence of ChatGPT-4 on augmenting students' learning outcomes concerning music. To this end, an experiment was conducted with 74 undergraduate students hailing from Chugye University for the Arts, Korea. The participants were randomly assigned to two distinct groups. One experimental group was assigned to utilize ChatGPT-4, while the control group received traditional teacher-led instructions. Additionally, a questionnaire was employed. Data analysis was conducted using SPSS 22. The results reveal that the experimental group exhibits a higher level of musical knowledge acquisition in comparison to the control group. Moreover, the results reveal that the experimental group demonstrates enhanced proficiency in the acquisition of musical knowledge when assessing their scores prior to and following the implementation of the intervention. The results indicate that the students demonstrate a propensity to utilize ChatGPT-4 as a tool for enhancing their music learning abilities. The influence of ChatGPT-4 on students' music learning outcomes is noteworthy, as it possesses the capacity to foster a perception of tranquility and ease among students. A multitude of suggestions and recommendations were proffered.

Keywords Artificial intelligence · chatGPT-4 · Music learning · Students

1 Introduction

Following the COVID-19 pandemic, the significance of higher-education's underlying components has increased exponentially throughout time (Abbass et al., 2022a, b, c; Abbas et al., 2022a, b, c). The contemporary focus of research in higher education is the quality of the education received by students (Abbass et al., 2022a, b, c; Abbas et al., 2022a, b, c). Moreover, there has been a shift in how people learn and teach thanks to various technological developments, such as smartphones, tablets, online courses, AI, and social media (Meng et al., 2023; Abbass et al., 2023, Abbass et al., 2019). Music education in higher institutions has received a considerable amount of interest from many researchers who are aiming to improving it (Esling & Devis, 2020; Klanjscek et al., 2021; Quan, 2019; Xie & Leung, 2011; Ho & Law, 2006). The reason for this lies in the fact that music education is helpful for students, inspiring new ideas and novel ways of thinking. It is an area of knowledge that allows some students to feel safe when experimenting with new things and creating original art (Klanjscek et al., 2021; Xie & Leung, 2011; Ho & Law, 2006). In contemporary times, the rapid progressions in the development of multimodal technology have precipitated substantial paradigm shifts within the domain of music (Gouzouasis & Bakan, 2011; Huang & Yu, 2021a, b). In recent years, there has been a notable proliferation of learning platforms and mobile applications that have integrated artificial intelligence technology (Rahmat et al., 2022). This is because music students have recently been able to enhance their learning experiences through the utilization of artificial intelligence technology (Klanjscek et al., 2021; Quan, 2019). This innovative approach has provided them with the opportunity to access a wide range of practice exercises, thereby expanding their repertoire of musical skills and knowledge (Esling & Devis, 2020; Klanjscek et al., 2021). These platforms and applications aim to offer personalized lessons, exercises, and feedback specifically designed to enhance students' musical skills (Barbosa, 2021; Ye, 2020). The aforementioned capability empowers individuals to iteratively enhance particular techniques and musical elements while avoiding the pitfalls of monotony (Esling & Devis, 2020; Klanjscek et al., 2021; Quan, 2019; Ye, 2020). These technological advancements facilitate the ability of applications to effectively process and respond to user inquiries, while also providing comprehensive evaluations pertaining to various training exercises (Xie et al., 2023).

Notwithstanding the advancements achieved through the utilization of artificial intelligence-driven systems and software, obstacles exist that have made it more difficult for educational institutions to effectively provide and distribute music education. Examples of significant contributors to problematic classroom environments include teachers' lack of expertise concerning subject matter and an inadequate availability of appropriate materials (Ho, 2014). Multiple factors, including an excessive emphasis on "core academic disciplines", such as language, literature, and mathematics, and the employment of unqualified music teachers are factors that contribute to the persistence of these problems. The research conducted by Xie and Leung (2011) illustrates the challenges teachers encounter when attempting to develop instructional activities within the context of music education. In their research, Ho and Law (2006) observed that teachers were dissatisfied with the quality and quantity of materials

available to their students. Hence, a notable proportion of students experience challenges during their musical acquisition process. The exposure to ineffective music materials can result in difficulties in increasing their knowledge and expanding their background in the field of music learning (Esling & Devis, 2020; Klanjscek et al., 2021; Quan, 2019). The insufficient involvement and dedication of students in being exposed to music have created a less than ideal educational setting, as music proficiency necessitates regular and consistent effort (Esling & Devis, 2020; Klanjscek et al., 2021; Quan, 2019).

Despite the emergence of several artificial intelligence systems, such as the recent innovation of OpenAI's ChatGPT-4, the aforementioned challenge remains prevalent in this field. ChatGPT-4 exhibits considerable potential within the realm of education and has garnered recognition in the literature for its capacity to support and enhance music education outcomes (McGee, 2023). An example of a potential instructional strategy involves facilitating student engagement with ChatGPT-4 through dialogs, which can serve to augment their understanding of complex music theory principles (Javaid et al., 2023; Li & Wang, 2023; McGee, 2023). This interactive approach encompasses a wide range of topics, including scales, chords, harmony, rhythm, and various other related concepts. The tool has the capacity to streamline and distill these complex concepts, thereby accommodating individuals who are in the early stages of their learning journey, as well as those who have progressed to an intermediate level of understanding (Javaid et al., 2023). Moreover, ChatGPT-4 has the capability to provide recommendations pertaining to instrumental technique, optimal finger placement, effective practice regimens, in addition to offering insights into the areas of sound manipulation, audio blending, and audio mastering. The user's text suggests that the system has the capability to suggest various plugins, effects, and methodologies to effectively attain the desired sonic outcomes.

The utilization of ChatGPT has garnered significant interest from researchers for various aims and scopes (Gilson et al., 2023; Javaid et al., 2023; Liu & Ma, 2023; Strzelecki, 2023; Warnerfjord, 2023). For instance, researchers in the field of English learning have taken notice of the growing use of ChatGPT-4 technology (Xiao and Zhi, 2023; Mizumoto and Eguchi, 2023; Peng et al., 2023; Schmidt-Fajlik, 2023). Previous research has shown that ChatGPT-4 uses a blend of unsupervised pre-training and supervised fine-tuning parameters to produce responses that mimic those of a human expert while answering queries and discussing topics (Dwivedi et al., 2023). Researchers have also proved that ChatGPT-4 technology excels at assisting users in acquiring appropriate information and promoting engaging conversations (Kohnke et al., 2023). Another researcher has investigated how ChatGPT-4 provides substantial educational advantages by simplifying assignment completion tasks, recommending scholarly articles, and empowering students with the capacity to address complex challenges through the provision of crucial knowledge and innovative problem-solving approaches (Zhai, 2022). However, the literature advocates some different results and perspectives concerning the effectiveness of ChatGPT-4 on student learning outcomes. Meanwhile, other studies have shown that the use of technology has negative impacts on students' education outcomes. Ajlouni et al. (2023) described the ChatGPT system as a conversational artificial intelligence chatbot in their study. This shows that it has the ability to generate text in varied contexts, such as formal,

casual, or creative writing scenarios, which present instructional issues. ChatGPT-4's easy information processing capability from a text input results in a lack of creativity in its output (Kasneci et al., 2023; Kohnke et al., 2023). However, if handled appropriately, these obstacles can ameliorate learning outcomes by introducing students to the social options, complexities, and hazards associated with artificial intelligence implementations. Xiao and Zhi (2023) demonstrated in their research that ChatGPT-4 creates answers by paraphrasing its sources without providing an attribution, which raises concerns of plagiarism. The model also results in inconsistent and unintelligible statements. The benefits and disadvantages of adopting ChatGPT technology in a higher-education setting are the topics addressed in Rawas' (2023) research. Personalization, interactivity, AI-powered tutoring, and content creation were highlighted as positive outcomes. On the contrary, obstacles included things like bias and ethics, a lack of human engagement, technical challenges, high costs, a lack of clarity in planning, and security and privacy issues. In their research, Ali et al. (2023) highlight that there are a number of potential problems that can occur as a result of using ChatGPT-4 technology in the classroom. Confidentiality and ethics are necessary factors to consider when discussing matters like copyright infringement and authorship questions. In this regard, it is worth noting that different studies yield diverse results, and further research is necessary to ascertain the impact of ChatGPT-4 technology on music education outcomes and whether ChatGPT-4 is beneficial for students.

However, its unexplored potential in the realm of music education presents a promising avenue for further investigations (McGee, 2023). Achieving a more complete understanding this topic is essential and beneficial as it is the first step in enabling students to expand their music knowledge and properly utilize transformative artificial intelligence technologies for music teaching purposes. Notwithstanding its use among learners in different fields on a global scale, a thorough comprehension of the impact of ChatGPT technology on students' music learning outcomes remains an issue that needs to be firmly established (McGee, 2023). Furthermore, it is worth noting that there is a notable number of empirical studies that provide substantial evidence supporting the advantages linked to the incorporation of ChatGPT-4 technology into various educational settings that focus on music learning (Li & Wang, 2023; McGee, 2023). In order to bridge these gaps identified in the research, the present study aims to investigate the potential influence of ChatGPT-4 technology on augmenting the quality of students' music learning experience.

This study offers valuable insights that can assist policymakers in making well-informed decisions regarding the integration of ChatGPT-4 technology as a supplementary tool to augment students' music education experience. These resources have the potential to be utilized in various educational settings, including both classroom and non-classroom environments. They are designed to align with the objectives and content of the formal music curriculum. This study demonstrates the potential for generating innovative research on the urgent matter of incorporating ChatGPT-4 technology in a music education setting, a topic of considerable importance for governments, educational institutions, teachers, and students globally. The recent discoveries in the literature not only bolster the existing research conclusions, but also enhance our understanding of ChatGPT-4's potential within the field of music education. The abovementioned implications have significant ramifications for the world-

wide integration of ChatGPT-4 technology in the field of education and the pursuit of musical knowledge.

2 Literature review

2.1 ChatGPT-4 in music education

The integration of technology into educational settings has made it possible to create educational resources that are both interesting and entertaining to use (Maqsood et al., 2021). It has been demonstrated in the research that using these tools can improve students' long-term memory retention of material and can be adapted to meet a variety of students' preferred modes of instruction (Yao et al., 2022; Maqsood et al., 2021). At present, it is possible, with the assistance of Internet resources, to provide students with individualized instructions tailored to their individual tastes and preferred methods of learning (Khalid et al., 2023; Rehman et al., 2023). People who have difficulty learning in the setting of a conventional classroom would benefit tremendously from this method (Rahmat et al., 2022; Meng et al., 2023). Because the Internet is so readily accessible, people from all walks of life are presently able to take advantage of the extensive variety of online learning opportunities made available to them by educational institutions located all over the world (Abbas et al., 2019, Abbas et al., 2022a, b, c). Many more people at present have the option to pursue higher education because of the democratization of education, compared to the past (Abbas et al., 2023).

The introduction of artificial intelligence into the field of education can have far-reaching effects, such as a radical shift in teaching strategies and classroom dynamics (Henriksen et al., 2021; Hoey et al., 2018). Artificial intelligence technologies, including chatbots and ChatGPT, show a lot of promise in enhancing the classroom learning experience (Javaid et al., 2023). ChatGPT is an AI-powered conversational agent that can answer questions, explain concepts, and create educational resources for its users (Javaid et al., 2023; Kasneci et al., 2023). The ChatGPT platform is based on the freely available GPT-3.5 model. The introduction of GPT-4 is a major improvement on previous versions of the ChatGPT Plus system. GPT-4's observable abilities include a great propensity for successfully interpreting and assessing many forms of inputs, including those in the visual and linguistic domains (Dwivedi et al., 2023). As a result of this, the GPT-4 system, at present, has the ability to produce insightful and meaningful textual responses to proposed questions. Across a wide range of professional and academic evaluation metrics, the system exhibits a performance that is on par with that of human experts in the field (Halaweh, 2023).

The ChatGPT-4 system's outstanding ability to absorb data and dynamically adapt its replies bodes well for the development of learning outcomes (Javaid et al., 2023). In their study, Mijwil et al. (2023) presented empirical results that demonstrated the potential of this phenomenon to improve the worldwide accessibility, efficiency, and efficacy of educational processes. Notable successes in a wide range of creative domains have been demonstrated in the research by the advances made in machine learning-based artificial intelligence technologies (Lupker & Turkel, 2021; Russell &

Norvig, 2020). These domains include content creation, information analysis, content enhancement, information extraction, information enhancement, and data compression. The concepts of natural language understanding, response generation in response to user cues, and conversational engagement displayed by the ChatGPT-4 system are quite outstanding. The benefits of utilizing the ChatGPT-4 system in the classroom have been studied by Baidoo-Anu and Owusu Ansah (2023). The evidence presented in their study suggests that students who use ChatGPT-4 technology can benefit from enhanced conversational abilities, increase their engagement with system interactions, and receive adequate academic support for their studies. The ChatGPT-4 system's adaptability to each student's specific demands has been noted by Rudolph et al. (2023). This is accomplished by thinking about certain ideas, like their prior experience and present skill level. In addition, it can be used to evaluate students' knowledge and skills. Using ChatGPT-4 technology, students have the ability to gain insights into their academic progress and accomplishments (Javaid et al., 2023). It also makes it easier for them to receive answers and responses, and provides them with the option to tailor the level of assistance they receive in order to successfully assess how well they are doing (Javaid et al., 2023; Kasneci et al., 2023). According to Cotton et al. (2023), students' interests can be piqued and distance learning made more manageable by incorporating ChatGPT-4 technology into the development of interactive and game-based assessments and problems.

A similar idea can be perceived in the potential improvements to students' music education outcomes with the use of the ChatGPT-4 system. Online software, like ChatGPT-4, provides music students with a flexible workspace that can be accessed from any device (McGee, 2023). The potential for artificial intelligence and ChatGPT-4 technology to facilitate more tailored and interesting educational interactions is exciting (Warnerfjord, 2023). The ability to tailor lessons to each student's unique objectives and musical preferences is one of ChatGPT-4's many strengths (Javaid et al., 2023; Adiguzel et al., 2023). By considering the learner's background, interests, and preferred learning style, ChatGPT-4 technology can create individualized music theory exercises, practice routines, and explanations. Having the flexibility to create changes makes for a more engaging and beneficial learning environment (Baidoo-Anu and Owusu Ansah, 2023; Hariri, 2023). The ChatGPT-4 system has the potential to act as a live, online, music instructor, guiding and critiquing students as they hone their music skills (Li & Wang, 2023). The system can be used to monitor students' instrumental performances, evaluate their technique, and offer helpful feedback for improvement (Li & Wang, 2023). Students' learning outcomes can be accelerated if they receive feedback promptly (Li & Wang, 2023; McGee, 2023). The ChatGPT-4 system can simplify difficult concepts in music theory by providing clear explanations and rapidly responding to any proposed questions. In response to the user input aspect, the ChatGPT-4 system can generate ideas, chord progressions, and lyrics that can be used by aspiring composers and songwriters. The aforementioned tool is priceless as a means of expression; it encourages artists to venture into uncharted creative territory.

In this regard, it can be stated that the changes in the traditional structure of music instruction have resulted from the introduction of artificial intelligence into this field (Gouzouasis & Bakan, 2011; Huang & Yu, 2021). Specifically, the incorporation of

computer music systems and cutting-edge, highly intelligent music software has led to significant improvements in the quality of music education as a whole (Barbosa, 2021; Li & Wang, 2023; Ye, 2020; Zhang, 2020a, b). Artificial intelligence has the potential to transform immaterial ideas into materialized forms because it is not bound by the constraints of time and space. Artificial intelligence has the potential to enrich the music teacher's workflow, captivate students' attention, and accelerate the learning process to coordinate their senses more effectively (Esling & Devis, 2020; Klanjscek et al., 2021; Quan, 2019).

In their study, Wei et al. (2022) argued that integrating artificial intelligence into the pedagogical framework of a network could provide teachers and students with greater control over the level of discourse in the classroom setting. The incorporation of artificial intelligence into the field of music education has had far-reaching effects on both the effectiveness of individual lessons and the overall music education system. Implementing an AI-driven, online, instructional platform into the context of music education for undergraduate music majors is one way to enhance the learning environment and make it easier to offer more individualized and specialized music courses. Teachers and students alike will have a better opportunity to embrace and profit from the online learning revolution if they have access to AI technology. This latter aspect offers a detailed example of the evaluation process that can help further music education outcomes. Teachers and students alike can benefit greatly from and be assured of the most efficient enhancement of the online instructional model by opting for the use of artificial intelligence rather than other traditional approaches. Hence, AI's potential to improve music education lies in its analysis of not just musical performances, but of the performer's behavior and facial expressions as well (Li & Ismail, 2022).

In reviewing the current literature on the ChatGPT-4 system, a considerable amount of attention has been given to the widespread effects of ChatGPT-4 technology in the field of education. The majority of studies conducted on this topic focus on the topic's potential effects on the areas of second language learning, research ethics, and classroom ethics. According to the research conducted by Baidoo-Anu and Owusu Ansah (2023), students who made use of ChatGPT-4 technology improved their linguistic competence level, interacted with the system more actively, and received sufficient scholastic support. These results are based on a survey of student users of ChatGPT-4 technology. In their study, Wei et al. (2022) also showed that educators and students alike could benefit from a greater classroom discourse regulation when AI was embedded into a network's pedagogical framework. Moreover, the empirical evidence offered by Mijwil et al. (2023) shows that this phenomenon can improve the accessibility, efficiency, and efficacy of educational processes all over the world. The application of ChatGPT-4 technology to EFL (English as a Foreign Language) writing/teaching domains was studied by Guo and Wang (2023). The results showed that ChatGPT-4 technology was a useful tool for providing feedback on students' work from multiple angles, including content, organization, and language. However, a significant gap in the literature still exists regarding how the ChatGPT-4 system contributes to students' musical development experiences. The relationship between these factors was not adequately studied. The current study aimed to address this research gap and explore the impact of ChatGPT-4 technology on students' learn-

ing experiences of music. Incorporating a group of Korean first-year students in this study who were enrolled in actual music classes helped accomplish this goal. This expanded the context in which our results could be useful. In this regard, our results contribute to the existing literature by providing a richer and more nuanced picture of the connection between the ChatGPT-4 system and students' musical education outcomes. This study not only fills a major knowledge gap in the research, but also offers important insights into those in charge of overseeing the expansion and development of music education.

2.2 Students' acceptance of ChatGPT-4 technology

In order to effectively optimize the efficacy of the ChatGPT-4 system as a useful tool for music education purposes, it is imperative to duly consider and prioritize students' responses and academic performances. This entails acknowledging and valuing their opinions and acceptance of the tool, thereby ensuring its alignment with the intended educational objectives. The application of ChatGPT-4 technology to fields, like higher education (Strzelecki, 2023), English language learning (Liu & Ma, 2023), and healthcare training (Gilson et al., 2023), dominates the current conversation in the research area. In their study, Lund and Wang (2023) interviewed ChatGPT users to learn more about the program's effects on the education sector. On the other hand, Perkins (2023) weighed the pros and cons of employing ChatGPT technology in fields as diverse as academia, publishing, authorship, and writing. The application of the ChatGPT system in K-12 and higher education has only recently emerged as an area of study concerning research and development. User acceptance of technology plays a critical role in promoting the spread and use of certain technologies (Davis, 1989). Students can be harmed if artificial intelligence fails to achieve widespread acceptance, as this can result in a surplus of artificial intelligence devices, the inefficient use of available resources, and a reduction in the rate of technological progress (Kirlidog & Kaynak, 2013; Lee & See, 2004).

The acceptance of ChatGPT technology among students has not been extensively examined by researchers, as it is a relatively new form of technology (Haglund, 2023). However, upon reviewing the existing literature, it is evident that numerous researchers have employed technology acceptance model to investigate the extent to which students accept artificial intelligence tools in the field of education. The technology acceptance model (TAM) was initially proposed by Davis et al. (1989) (Chen & Tseng, 2012). The TAM has consistently been the most widely utilized framework for illustrating the acceptance of technology within the field of education (Gomez-Ramirez et al., 2019). The proposed model posits that the adoption of technology tools is influenced by several factors, namely, perceived usefulness, perceived ease of use, attitudes, and intentions of use. The proliferation of computer technology in the 21st century has rendered it a ubiquitous and essential tool for students, enabling them to take charge of their own learning experiences in order to attain long-term educational objectives, such as acquiring proficiency (Lee et al., 2017). The primary objective of the TAM is to elucidate the key determinants that influence user behavior in relation to the acceptance of technology (Davis, 1989).

The determination of acceptance is influenced by various factors. The technology acceptance model comprises two primary components, namely, perceived ease of use and perceived usefulness. The concept of perceived ease of use pertains to an individual's perception regarding the level of effortlessness associated with utilizing a specific system. Perceived ease of use is predicated on the idea that people act in a way that benefits themselves and others because they believe it is easy to do this (Davis, 1989). The concept of perceived usefulness pertains to an individual's perception of the extent to which employing a specific system enhances their job performance outcomes. Recent research conducted in the field reveals that individuals who have positive beliefs about the effectiveness of ChatGPT-4 technology are more likely to engage in activities involving the use of this AI-powered technology for English language learning purposes outside conventional academic settings (Liu & Ma, 2023). In addition, attitude is another relevant component, which is defined as the acceptance or rejection of a system (Davis, 1989). The literature advocates that students expressed a favorable attitude towards the integration of ChatGPT technology in an education setting (Ofosu-Ampong et al., 2023). As teachers increasingly assume responsibility and demonstrate vested attitudes concerning the integration of technology in a classroom setting, confidence is cultivated, enthusiasm is enhanced, and apprehensions regarding its utilization become more prevalent (Xie et al., 2023). Hence, an individual's attitude plays a crucial role in determining his or her efficacy and interest in sustaining the use of a certain form of technology (Xie et al., 2023). Furthermore, the concept of intention to use refers to the objective, aim, or strategy of utilizing artificial intelligence or acquiring further knowledge about it (Lee, 2020). The acceptance of artificial intelligence depends on people's natural propensity to continue using technology following a positive experience. This means that people's habits have a significant impact on how they react to the introduction of new technologies (Alharbi & Drew, 2014). Individuals' subsequent actions are strongly correlated with the idea of intent to use (Xie et al., 2023). Several variables affect a user's likelihood of performing a given action. In numerous fields, including education (Xie et al., 2023; Tsai, 2014), researchers have experienced success when using the technology acceptance model to measure users' attitudes toward and intentions regarding new technologies. To summarize, the current research gaps concerning the acceptance of ChatGPT-4 technology in an education setting emphasizes the need to assess students' acceptance of the ChatGPT-4 system, specifically within the context of music instruction. Previous research demonstrate that multiple elements that can influence the adoption of technology in the field of education exist. The technological acceptance model (Davis, 1986) examines these aspects. We decided to investigate these specific concerns related to ChatGPT-4 technology in a music education setting, because of its widespread utilization in the Koran setting (Han and Sa, 2021; Kim & Lee, 2016; Lee et al., 2009), and its substantial market dominance in the study period.

2.3 The present study

This study aims to investigate the potential influence of ChatGPT-4 technology on augmenting the quality of students' music learning experiences. Additionally, this study aims to assess the effects of ChatGPT-4 technology on students' learning expe-

riences of music and their acceptance of the ChatGPT-4 model for improving their music learning outcomes. To achieve this, the following research questions were formulated:

1. Are statistically significant differences evident in the music learning experiences between the experimental and control groups following the implementation of ChatGPT-4 technology?
2. Is there a significant difference in music learning experiences prior and subsequent to the use of ChatGPT-4 technology?
3. Would students' acceptance of using ChatGPT-4 technology for improving their music learning experiences significantly change following the implementation of the ChatGPT-4 system?

3 Methodology

3.1 Research design

The primary objective of this study was to investigate the effect of ChatGPT-4 technology on improving the quality of students' music learning experiences. This study employed a quantitative approach based on experiment and questionnaire research methods. The adoption of quantitative research allowed for the collection of data from a diverse group of participants, facilitating the exploration and measurement of interactions between the relevant factors. It is generally agreed that numerical data, such as those gathered from experiments and questionnaires, are more objective than other types of data (Shah et al., 2023). The use of these methods helps to reduce the risk of bias that occur as a result of different people's perceptions of the data (Abunamous et al., 2022). In another study, it was revealed that uncovering relevant patterns, connections, and new insights would have been impossible to achieve without the use of quantitative data as well as the use of a statistical analysis (Abbas et al., 2024). Due to the uniformity of quantitative data, comparisons can be performed between various studies, people, and time periods (Xu and Boudouaia., 2023). These comparisons can be beneficial to the research. When making decisions about the aspects of corporate strategy and public policy, the use of quantitative data may be extremely helpful (Abbas et al., 2024; Shah et al., 2023). The use of empirical evidence makes it possible to conduct objective evaluations, free from the bias that anecdotes and individual preferences might exert (Peng et al., 2023; Rahmat et al., 2022). Quantitative approaches, particularly those used for collecting and evaluating relevant data, are more efficient in terms of both time and money, when compared to qualitative methods (Abbas et al., 2024; Peng et al., 2023; AL-Qadri et al., 2022). When compared to qualitative data, quantitative data can be perceived to enable the measurement of changes over time as well as the observation of trends, both of which can otherwise remain unnoticed (Bariu et al., 2022).

Experiments allow for an extremely exact control of variables in order to investigate how these elements affect the final result. It takes work to successfully man-

age and maintain an effective cause-and-effect relationship (Boudouaia et al., 2022). When a questionnaire is standardized, it ensures that the same questions are posed to all participants, and they are asked the questions in the same order (AL-Qadri et al., 2021; Meng et al., 2023). Because of this uniformity, any disparities in the data are demonstrated to be the product of genuine variances among the respondents rather than faults in the data collection process itself (Abbas et al., 2023; Xie et al., 2023). Because the results may be applied to a greater number of situations, thanks to the capability of scaling them up, their potential value in the real world may be increased (Xie et al., 2023).

3.2 Participants

The researcher's initial expectations for a large sample of students and institutions that would participate in the study did not materialize. Chugye University for the Arts, which is highly regarded in Korea, was the site of the research. Until the research team received approval from the institution's administration, it was unable to commence the study. In addition, a teacher who observed the significance of the need to harmonize the course provided their wholehearted support and approval. Seventy-four students enrolled in the first year of the music program at Chugye University for the Arts were polled. The participants were informed of the goals of the study and afforded the opportunity to provide their informed consent prior to any data collection activity. The primary objective of this research was to examine whether or not undergraduate students benefit more from ChatGPT-4 technology when it comes to the improvement of their music learning experiences.

3.3 Experimental procedures

The present study used a quasi-experimental design to investigate the effect of ChatGPT-4 technology on students' music learning experiences. The participants were randomly divided into 2 groups: experimental ($n=36$) and control ($n=38$). The experimental group used the ChatGPT-4 system in the learning process, whereas the control group did not. The two groups were led by the same teacher, who consistently employed the same teaching methodology in both experimental conditions. The teacher ensured that all the students received the same instructional materials and instructional directions across all classes. However, the ChatGPT-4 system was used exclusively in the experimental condition.

The investigation was conducted over a duration of 10 weeks in the spring semester of the academic year (2022–2023). The assessment of the students' music learning experiences was conducted through a pre-test administered during the initial week of the course for both groups. The introduction of ChatGPT-4 technology was accompanied by comprehensive instructions provided to the experimental group students, aiming to assist them in maximizing their utilization of the system. The implementation of the ChatGPT-4 system occurred during the second week of the intervention period. The students actively engaged in the process of elucidating novel concepts, genres, scales, chords, harmony, and composition techniques. Subsequently, they received feedback during the subsequent week and proceeded to refine their work

within a span of two weeks. Initially, the students acquired knowledge autonomously, without the assistance of the ChatGPT-4 system. Following the initial incident, ChatGPT-4 technology made valuable contributions towards the identification and resolution of the issues that emerged. The teacher provided elucidations regarding any inaccuracies and proffered justifications for any exclusions. The revision process was facilitated by the active engagement of students, who effectively integrated the constructive feedback offered by the ChatGPT-4 system. The proposed modifications were assessed and subsequently incorporated in relevant instances by the researchers. It was imperative to acknowledge that the researchers effectively utilized the pre-test problems as a valuable indicator of the foremost issues pertaining to music learning outcomes that necessitated the attention to and resolution of the subsequent meeting. In order to attain this outcome, a comprehensive analysis was conducted on the students' weekly progress. To dissuade the practice of recycling, a post-test was administered subsequent to the initial intervention, incorporating novel content.

3.4 Tests

Two tests were conducted in the present study. The preliminary assessment, conducted during the initial week, was a diagnostic evaluation aimed at discerning students' background knowledge of music. This was achieved by posing inquiries pertaining to their familiarity and comprehension of some concepts, genres, scales, chords, harmony, and composition techniques. The participants independently wrote scholarly essays without receiving any guidance from an instructor, peers, or online resources. The second assessment, referred to as the post-test, necessitated students to articulate their acquired knowledge from the experiment by offering detailed explanations and supporting their claims with illustrative examples to demonstrate their comprehension. The objective of the study was to improve upon the initial test results we obtained and revolved around the creation of a comprehensive study schedule for a specific academic program. The post-test aimed to assess the impact of the ChatGPT-4 system on the students' acquisition of musical knowledge.

3.5 Questionnaire

The researchers utilized the artificial intelligence acceptance instrument from the original (Venkatesh & Davis, 2000) and the extended technology acceptance model scale (Sánchez & Hueros, 2010) to measure the acceptance of artificial intelligence in this study. Additionally, perceived ease of use, perceived usefulness, attitudes, and intention to use were all factors that were considered in the study. The initial survey comprised a total of 15 items. The questionnaire items were assessed using a 5-point Likert scale, ranging from 5 to 1, with the corresponding scores assigned as follows: strongly agree=5, agree=4, neutral=3, disagree=2, and strongly disagree=1. In this study, the researchers conducted an analysis on the study's instrument, which consisted of a total of 15 items. The objective was to thoroughly investigate the structure of the factors within the sample population. Prior to implementation, the surface validity of each questionnaire item was assessed by four experts who provided feedback, who agreed on the questionnaire contents. The study results reveal that the

Cronbach's alpha (α) values are 0.812, 0.843, 0.827, and 0.819. These results indicate that the reliability values are deemed acceptable (Peng et al., 2023; Xie et al., 2023).

3.6 Data analysis

The researcher systematically classified and quantified the participants' responses to the test questions using a 5-point scale. The upper limit of the scoring scale for participants in this study was set at "20," which corresponded to the degree of accuracy exhibited in their responses. The scoring process involved the participation of three raters. The raters engaged in a discussion to address any significant discrepancies in the assessment process, if such discrepancies existed. Cohen's kappa coefficient was utilized to assess the interrater reliability of the post-test assessment, and the analytical results we obtained demonstrated a high level of reliability, exceeding 0.80. In addition, the process of analyzing the data was executed through the utilization of Statistical Product and Service Solutions (SPSS) Software (Version 22.0). Given the significance of the data entry in this study, this was executed with meticulous care to ensure the acquisition of accurate outcomes. Ultimately, an analysis of the data was performed, followed by an interpretation of the results.

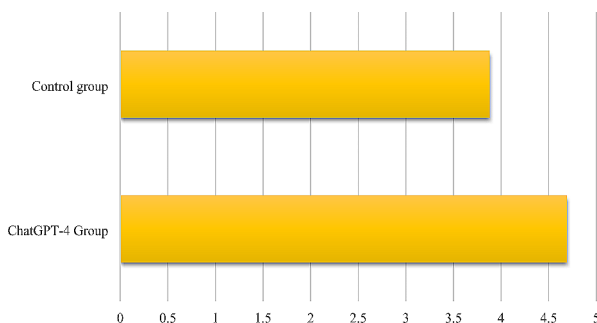
4 Results

Regarding the initial research question, which examined the potential existence of statistically significant differences in music learning outcomes between the experimental and control groups subsequent to the integration of ChatGPT-4 technology, a statistical examination utilizing the independent-sample t-test was conducted. The research aim encompassed a comprehensive examination of students' cognitive grasp and comprehension of fundamental concepts, genres, scales, chords, harmony, and composition techniques pertaining to music. The study's results were collected by employing a T-test. As shown in Table 1; Fig. 1, upon the successful implementa-

Table 1 Music learning results of experimental and control groups

	N	M	SD	<i>F</i>	<i>t</i>	<i>Sig</i>
ChatGPT-4 group	36	4.69	8.03	16.46	9.71	0.000*
Control group	38	3.88	8.61	0.040		

Fig. 1 Music learning results of students in experimental and control groups



tion of the intervention, the experimental group demonstrates a statistically significant increase in its mean score ($M=4.69$, $SD=8.03$) compared to the control group ($M=3.88$, $SD=8.61$). This difference was observed to be statistically significant at the 0.05 probability level ($p=0.000$). The results of this study suggest that there is a statistically significant difference in the post-test scores observed between the experimental group, which received the ChatGPT-4 treatment, and the control group, in relation to the acquisition of music knowledge. This suggests that the experimental group demonstrates advancements in their music learning outcomes in comparison to the control group as a result of utilizing the ChatGPT-4 system.

Regarding the second research question, which pertained to the potential differences in music learning experiences prior and subsequent to the implementation of the ChatGPT-4 system, a paired-sample t-test was employed. Based on the results presented in Table 2; Fig. 2, it is evident that the students belonging to the experimental group demonstrate enhanced music learning outcomes during the post-test ($M=4.69$, $SD=8.03$), compared to their pre-test scores ($M=3.11$, $SD=5.54$). The present study revealed a statistically significant difference at the 0.05 level of significance ($p=0.000$), suggesting a significant difference between the pre- and post-test scores in relation to music learning experiences following the implementation of the ChatGPT-4 intervention for students in the experimental group.

Regarding the third research question, which pertained to the potential alteration in students' acceptance of utilizing the ChatGPT-4 system as a means to enhance their music learning experiences subsequent to the implementation of the ChatGPT-4 system, t-tests were performed for the analysis. Fluctuations in the perception scores within the experimental group were evident through the implementation of the t-test. Table 3; Fig. 3 present the mean scores attained by the students in both the pre- and post-test phases, specifically focusing on the various components within four distinct sub-dimensions. These sub-dimensions are identified as perceived usefulness, perceived ease of use, attitude towards using, and intention to use.

Table 2 Experimental group results for pre- and post-test music learning experiences

	N	M	SD	95% Confidence interval of the difference		<i>t</i>	<i>Sig</i>
				Lower	Upper		
pre-test	36	3.11	5.54	5.83	12.22	7.56	0.170
post-test	38	4.69	8.03			9.71	0.000

Fig. 2 Experimental group results of music learning in pre- and post-tests

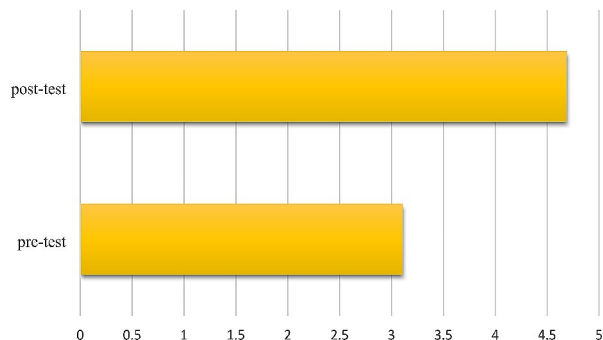
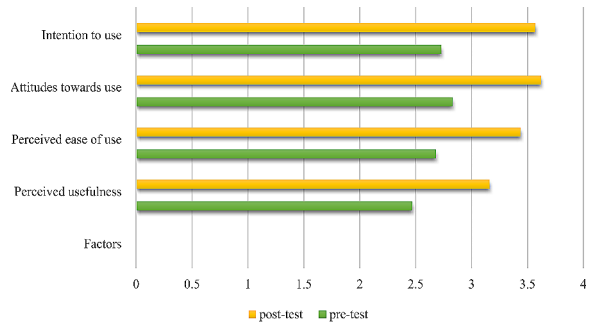


Table 3 Student acceptance results for using ChatGPT-4 technology in music learning practices

Factors		Pre-Test		Post-Test		<i>t</i>	<i>P</i> -value
		M	SD	M	SD		
ChatGPT-4 group	Perceived usefulness	2.47	4.16	3.16	4.31	2.46	0.001
	Perceived ease of use	2.68	4.84	3.44	4.76	2.53	0.020
	Attitudes towards use	2.83	4.40	3.62	4.51	2.49	0.001
	Intention to use	2.73	4.51	3.57	4.64	2.51	0.000

Fig. 3 Student acceptance results for using ChatGPT-4 technology in music learning practices

The results of the t-test analysis indicate that there are statistically significant improvements in the experimental group's perceptions of perceived usefulness, "perceived ease of use, attitudes towards use, and intention to use. The results of the study indicate that there is a significant increase in the post-test scores compared to the pre-test scores for the variables under investigation. The post-test scores for these variables were observed to be significantly higher than the pre-test scores ($t=2.46$, $p=0.001<0.05$; $t=2.53$, $p=0.020<0.05$; $t=2.49$, $p=0.001<0.05$; $t=2.51$, $p=0.000<0.05$, respectively). These p -values, all less than the predetermined significance level of 0.05, suggest that the observed differences in the scores are unlikely to have occurred by chance. The results of the study indicate that the mean values observed in the post-test are higher compared to those in the pre-test for the constructs of perceived usefulness, perceived ease of use, attitude towards using, and intention to use after the participants utilized the ChatGPT-4 system. Specifically, the mean score for perceived usefulness increased from 2.47 in the pre-test to 3.16 in the post-test. Similarly, the mean score for perceived ease of use increased from 2.68 in the pre-test to 3.44 in the post-test. Additionally, attitudes towards use presented an increase from a mean score of 2.83 in the pre-test to 3.62 in the post-test. Lastly, the mean score for intention to use increased from 2.73 in the pre-test to 3.57 in the post-test. The results indicate a notable change in students' perceptions regarding the efficacy of the ChatGPT-4 system in augmenting their music education experiences subsequent to its integration.

5 Discussion

The main aim of this research was to investigate the effects of incorporating the ChatGPT-4 system into the process of learning music. Specifically, the study aimed to assess the extent to which ChatGPT-4 technology may enhance students' music proficiency levels and their inclination to utilize the ChatGPT-4 system as a means of augmenting their music learning experience.

The results of the investigation show that the cohort subjected to the experimental conditions exhibit a more pronounced learning outcome concerning musical knowledge in comparison to the cohort assigned to the control conditions, as substantiated by their elevated average score. The identified difference was observed to possess a statistically significant effect. This result implies that the incorporation of the ChatGPT-4 system has a discernible influence on students' acquisition of musical knowledge, as evidenced by the experimental group's display of improved learning outcomes. The presented results demonstrate a striking similarity to the research inquiry undertaken by McGee (2023), which showed that the ChatGPT-4 system was able to successfully generate a musical score for a taekwondo tournament routine, thus providing evidence for its efficacy in facilitating students' acquisition of musical skills. In a similar vein, Zhang (2023) discovered that a considerable number of pupils exhibited a preference for intelligent piano instruction in a study. Furthermore, the study conducted by Li and Wang (2023) revealed that students who utilized the provided application demonstrated a superior academic performance compared to their counterparts enrolled in traditional classroom settings. In general, the utilization of artificial intelligence yielded a notable enhancement of 15% in their academic performance. However, the degree of training intensity exhibited variabilities, which consequently influenced the outcomes. The application yielded the following improvements in the scores: piano playing increased by 6.51%, solfeggio and music literature improved by 4%, and vocal signing had a modest increase of 0.56%. The incorporation of chatbots into a music instruction setting presents novel opportunities for advancements in the field. Therefore, the utilization of artificial intelligence, which possesses the ability to analyze music performances, as well as the behavior and facial expressions of the performer, presents teachers with a viable means to effectively instruct musicians with the assistance of artificial intelligence (Li & Ismail, 2022).

The study's results suggest that ChatGPT-4 technology has the capacity to serve as a beneficial resource for students seeking to enhance their music education. The potential benefits that may arise from incorporating the ChatGPT-4 system into music education offer instructors a stimulating incentive to reflect on the outcomes. The results derived from the experimental group offer persuasive evidence that the implementation of the ChatGPT-4 system in this setting has the potential to customize musical experiences to suit the specific requirements of each student. The enhancement of music learning outcomes can be optimized by employing personalized feedback and customized practice opportunities. These strategies enable instructors and students to concentrate on particular areas that require improvements.

Moreover, the results indicate that the participants in the experimental group demonstrate an enhanced proficiency in the music learning process when comparing their

scores for the pre- and post-tests. The observed discrepancy was determined to possess a statistical significance. The results of this investigation exhibit a resemblance to the scholarly inquiry conducted by Li (2023), which revealed a statistically significant increase of 24% in the likelihood of genre recognition among the participants following exposure to the technology. In contrast, it was observed that students pursuing non-music majors demonstrated a more substantial enhancement in their linguistic aptitude skills, with a notable increase of 27%. The study's results suggest a positive correlation between a modest level of musical knowledge among music students and improved literacy skills in music ($k=0.82$). Moreover, the results suggest that individuals who are not pursuing an education in music experience a significant improvement in their memory skills ($k=0.84$) and demonstrate enhanced musical literacy as a result of their knowledge acquisition. The results of a study conducted by Cui (2022) reveal that the incorporation of artificial intelligence into the experimental design yields noteworthy enhancements in the students' mastery of musical terminology literacy. Furthermore, the results of the study indicate that a significant majority of the participants, precisely 89%, express a notable degree of progress in the specified domain. Moreover, it is worth noting that a significant majority of the students, specifically 83%, demonstrate notable improvements in their proficiency for reading sheet music and their capacity to independently engage with musical material. The results of this study indicate that a substantial proportion of students, precisely 70%, exhibit a commendable level of competence in the area of piano performance, thereby enabling them to proficiently execute a musical composition. Furthermore, it is worth noting that a significantly higher percentage, specifically 90%, of the student population demonstrated a notable level of technical proficiency that aligned with the skill set typically observed in highly skilled pianists.

The integration of artificial intelligence within the realm of music education yielded substantial transformations to the traditional methodology. Significantly, the integration of computer music systems and sophisticated music software endowed with advanced levels of intelligence yielded considerable improvements to the overarching caliber of music education (Zhang & Wan, 2020a, b). Moreover, the incorporation of artificial intelligence technology, which includes the utilization of three-dimensional sound effects, real-time sound effects, and virtual three-dimensional scenes, holds promise in enhancing the effectiveness and efficiency of the educational experience concerning music (Huang & Yu, 2021a, b). The amalgamation of artificial intelligence with big data intelligence has the capacity to replicate the cognitive processes inherent in human thinking and reasoning (Rahmat et al., 2022; NeJhaddadgar et al., 2022). The integration we mention in this study makes it easier to effectively improve and refine the algorithms that are utilized in neural networks (Khalid et al., 2023; Abbas et al., 2019). The incorporation of artificial intelligence in the field of music education holds significant promise as a potent and valuable instrument for facilitating the learning processes of students (Huang, 2021a, b).

The results obtained by this study have significant implications for individuals working in the field of music education, including educators and people responsible for creating the curricula. The results show that integrating the ChatGPT-4 system with classroom instructions can help students learn music more effectively. Hence, music teachers and curricular developers can include lessons on digital literacy in

their curricula. As a result of its flexibility, the ChatGPT-4 system can be used to provide students with individualized music education programs. Moreover, its capacity to deliver rapid responses further improves its potential in this domain. The benefits of incorporating the ChatGPT-4 system into music instructions are substantial, as it paves the way for a more inclusive classroom and encourages student engagement and collaboration. Students must acquire and refine their digital literacy skills to ensure that AI-driven solutions are used to their full potential.

Additionally, the results obtained by this study indicate that students demonstrate a willingness to utilize ChatGPT-4 technology as a means of enhancing their learning of music. The experimental group exhibited noteworthy enhancements in students' perceptions of perceived usefulness, perceived ease of use, attitudes towards use, and intention to use. The post-test scores exhibited a statistically significant increase in comparison to the pre-test scores for the aforementioned variables. The results presented herein exhibit a notable resemblance to the aforementioned investigation conducted by Strzelecki's (2023) that the outcomes provided substantiation for a significant majority, specifically nine out of ten, of the proposed theories. The results of the study indicate that habit emerges as the primary predictor of behavioral intention, with performance expectancy and hedonic motivation also playing notable roles. The study revealed that behavioral intention emerged as the foremost determinant of use behavior, followed by personal innovativeness as a subsequent predictor. Moreover, the empirical investigation conducted by Foroughi et al. (2023) revealed noteworthy results regarding the impact of various factors on individuals' inclinations to employ the ChatGPT-4 system. Specifically, their study highlighted the significant influences of performance expectation, effort expectancy, hedonic motivation, and learning value on the aforementioned desire. Strzelecki (2023) and Foroughi et al. (2023) based their research on the Unified Theory of Acceptance and Use of Technology (UTAUT2) framework developed by Venkatesh et al. (2012), it is worth mentioning that the current study did not employ this specific theoretical framework. However, the results of the aforementioned studies are consistent with the outcomes observed in the present inquiry. The current results align with the research conducted by Liu and Ma (2023), demonstrating that, while perceived ease of use does not have a direct impact on learners' attitudes towards the use of the ChatGPT-4 system, it does exert an indirect influence on their attitudes by means of the mediator, perceived usefulness. Furthermore, a notable result was obtained indicating that individuals who possessed favorable perceptions regarding the efficacy of the ChatGPT-4 system demonstrated a heightened propensity to partake in activities that entailed the utilization of this AI-powered technology for the purpose of English language acquisition beyond traditional academic environments. The observed inclination demonstrated a strong and favorable indication of their subsequent engagement with ChatGPT-4 technology. Ofosu-Ampong et al. (2023) conducted a study where they observed the attitudes of participants towards the integration of artificial intelligence in the field of education. The results reveal that a significant majority of participants ($n=102$, 69.9%) express a favorable attitude towards this integration. In contrast, it is worth noting that a smaller percentage of individuals involved in the study ($n=44$, 30.1%) exhibited a dissenting perspective regarding the integration of artificial intelligence in the field of education.

The empirical evidence indicates that there notable improvements can be noticed in the students' subjective assessments of the characteristics referred to as perceived usefulness and perceived ease of use. The study revealed that students perceived the ChatGPT-4 system as having qualities that made it easily accessible and beneficial for fulfilling their individual needs. These results emphasize crucial insights for policy makers to reassess the implementation of appropriate policies that can successfully support the acceptance and integration of ChatGPT-4 technology in music educational settings. These results also emphasize the significance of prioritizing the advancement of user-friendly and effective educational technologies, specifically ChatGPT-4, in order to provide positive experiences for the students using it. The research results suggest that teachers can benefit from actively engaging in collaborative efforts with ChatGPT-4 to enhance the overall efficacy of music education outcomes. The noticeable increase in students' positive attitudes towards the implementation of the ChatGPT-4 system in the music education setting and their stated dedication to utilizing it highlight the encouraging potential of ChatGPT-4 technology. This result indicates that students should have a propensity to adopt and utilize ChatGPT-4 technology as a method to meet their music educational needs. However, the positive outcomes have the potential to stimulate additional academic investigations exploring the possible utilization of ChatGPT-4 technology in other educational environments. ChatGPT-4 technology has the capacity to improve educational practices by providing assistance to students across a broad spectrum of academic disciplines. This has the capacity to enhance the accessibility and efficacy of education. When considering the potential benefits of AI, it is important to view it as a supplementary tool rather than a substitute for human teachers.

6 Conclusion

The results of this study suggest that the ChatGPT-4 system shows great potential as an effective tool for improving students' music learning experiences, and fostering a positive attitude and acceptance among students. Students' musical knowledge and abilities were improved via the usage of ChatGPT-4 technology, and they also became more receptive to and comfortable with the use of ChatGPT-4 in the classroom as a result of this. These results indicate the significance and utility of ChatGPT-4 for use in a classroom setting, particularly for the purpose of learning music. Following the introduction of ChatGPT-4 technology into the music course, it was observed that, not only did students' musical literacy levels improve, but so did their acceptance of its use and level of passion. The benefits of using ChatGPT-4 technology were not only statistically proven, but they were also apparent in comparisons of evaluations, which showed that it could considerably improve the outcomes of music learning. It can be stated that ChatGPT-4 technology can address existing issues and fill in the knowledge gaps for students concerning a wide variety of musical themes. Because it can be accessed at any time, students can use ChatGPT-4 technology to study and hone their musical skills even when they cannot attend school. Students can benefit from the ChatGPT-4 system's ability to simplify difficult musical concepts and elaborate on them in great depth.

6.1 Policy recommendations

This study proves that the ChatGPT-4 system possesses substantial potential for enhancing students' experiences of learning music, thereby necessitating governments to duly acknowledge and consider this role. The educational system is very sensitive to monetary and policy decisions made at the national level. It is recommended that governments should ease the process of incorporating the ChatGPT-4 system into educational systems if they acknowledge the potential of ChatGPT-4 technology in this setting. It is important for decision makers who are considering using the use of ChatGPT-4 technology in a music education setting to be familiar with both the platform's advantages and disadvantages, as well as the prerequisites necessary for using the system. It is possible that preliminary research and test projects will be required in order to discover the most productive applications of the ChatGPT-4 system in this field. In addition, policies should be developed to ensure that the ChatGPT-4 system complements human instructors, rather than replacing them. These policies should address problems, such as access equity, data privacy, and similar topics. The significance of policymakers' considerations of this aspect lies in the facilitation of a seamless integration of the ChatGPT-4 system into the established music education curricula. This involves making ChatGPT-4 technology an integral part of teaching and learning processes, rather than merely using it as an extra resource. Redesigning lessons, providing instructors with appropriate technology use training, and expanding access to technology are all possible methods to achieve this goal. The provision of professional development opportunities can be strategically implemented to support teachers in effectively integrating artificial intelligence tools, such as ChatGPT-4 technology, into their instructional methodologies within a classroom setting. An avenue worth exploring involves examining the allocation of resources towards the examination, development, and implementation of the ChatGPT-4 system within the context of music learning.

6.2 Implications

Based on the results of this study, it is implied that teachers should devise a comprehensive integration strategy in order to maximize the advantages of the use of ChatGPT-4 technology in a music learning setting. This strategy should be carefully planned, considering the individual requirements and needs of each student in terms of their music learning experience, as well as the overarching objectives of the curricular and educational institutions. Sustaining the effectiveness of the intervention necessitates teachers to perform ongoing evaluations and refinements of the use of the ChatGPT-4 system for music learning purposes. This approach may help tutors diagnose the needs of students and simultaneously improve the use of the ChatGPT-4 system in ways that assist in the improvement of students' music learning experiences. In order to enhance the pedagogical approach to the learning of music, it is recommended that teachers consider using ChatGPT-4 technology as an auxiliary tool. Student preference for the use of ChatGPT-4 technology in the context of music education appears to be supported by our study. The students rated, in general, all four elements comprising the technology acceptance model favorably

(perceived usefulness, perceived ease of use, attitudes, and behavioral intentions). The students' acceptance of the use of ChatGPT-4 technology for music education purposes appeared to be significantly impacted by certain attributes, such as perceived usefulness, perceived ease of use, attitudes, and behavioral intentions. Hence, it is imperative that educators develop music-oriented instructional modules that are compatible with ChatGPT-4 technology. The students' enthusiasm for utilizing the ChatGPT-4 system in a classroom setting can be significantly piqued in this manner. Regarding the students who demonstrated exceptional utilization capabilities of the ChatGPT-4 system during music learning practices, the educators ought to contemplate the implementation of incentive systems as a means of effectively recognizing and motivating them. Additionally, instructors ought to provide support to students who have had negative experiences when using the ChatGPT-4 system. Irrespective of the potential and functionalities of ChatGPT-4 technology, educational institutions ought to extend invitations to technology specialists to deliver lectures to their students. This would increase students' understanding of the practical features of the ChatGPT-4 system.

Moreover, the potential establishment of collaborative partnerships between developers of the ChatGPT-4 system and teachers is promising for the development of personalized instructional resources enhanced by the utilization of ChatGPT-4 technology. By providing developers with regular feedback from educators using the ChatGPT-4 system in educational settings, educators can help developers boost the AI's efficacy and ensure it meets the needs of educational settings. Because of their first-hand familiarity with the technology, as well as their understanding of students' requirements and preferred modes of learning, teachers are in a unique position to offer valuable insights into how the ChatGPT-4 system can be used to achieve the greatest possible impact in the classroom. However, ChatGPT-4 developers can use teachers' suggestions to address the challenges in the classroom by utilizing their technical understanding, as well as their creative and innovative problem-solving talents. Teachers and programmers have the ability to produce superior pedagogical tools by utilizing the advanced capabilities of the ChatGPT-4 system. This addressed interesting and useful alternatives to more traditional approaches, such as interactive learning materials, adaptive exams, and support systems. Together, teachers, ChatGPT-4 developers, and researchers can provide new solutions to the long-standing problems in a music education setting, such as narrowing the achievement gap, assisting students who have special needs, and streamlining the management of large classrooms full of diverse students.

Based on the study results, it is implied that students who are interested in the fields of music composition and production will gain a great deal of knowledge and experience from the opportunity of becoming familiar with a diverse selection of musical styles, forms, and concepts by using the ChatGPT-4 system. This will be very beneficial to them. Students benefit from having access to an interesting and cutting-edge platform, like ChatGPT-4, which enhances their overall music educational experience. It is possible that the power of the ChatGPT-4 system to deliver real-time feedback, answer queries, and provide personalized learning routes will make the process of music learning more approachable and pleasurable. The personalized tools and exercises that the ChatGPT-4 system can provide are beneficial to the study of

music in a variety of contexts, including theoretical comprehension, understanding composing approaches, and the investigation of musical genres. Because of the proliferation of the ChatGPT-4 system, students, at present, have access to a plethora of information, which may help strengthen their capacity for creative thought. Students who have a great interest in the creation and production of music may derive a great deal of value from the options provided by the ChatGPT-4 system.

Confirming this study would provide a solid basis for the future research conducted on the ChatGPT-4 system in the field of music education and how it is perceived by college students. Recent research studies primarily focus on analyzing the effects of the ChatGPT-4 system on language learning outcomes and exploring ethical concerns in educational environments, all within a specific time frame. By employing a longitudinal strategy, researchers can effectively observe and comprehend any changes in the impact of the ChatGPT-4 system on students' music learning experiences. In addition, this approach allows for the successful examination of the acceptance and longitudinal use of the ChatGPT-4 system in an education setting.

6.3 Limitations and future research directions

The research conducted in the field of education certainly exhibits flaws, and this must be recognized in the literature. The study's reliance on the ChatGPT-4 system as its major research instrument raises some concerns about the generalizability of the results. Future study efforts will benefit from performing comparative analyses involving a broader set of artificial intelligence technologies in addition to ChatGPT-4 technology and teachers' instructions. Discovering what works best for individual students can lay the groundwork for a successful learning environment. The primary goal of this research was to examine the usefulness of the ChatGPT-4 system as a tool for improving students' music education experiences. Furthermore, it should be noted that the study was only completed for a total of ten weeks. Therefore, future research projects would be improved by adopting a longitudinal technique, which would allow for a more thorough study of students' opinions over time. This would improve the students' in-depth comprehension of the implications of incorporating inputs from multiple sources on their perspectives.

Funding This research did not receive any funding.

Data availability The data can be made available upon request.

Declarations

Informed consent Informed consent was obtained from all the students participating in this study. Confidentiality was maintained by not requesting names or any other information that would identify the students involved. The subjects were informed of their rights to withdraw from the investigation at any time.

Conflicts of interest No potential conflicts of interest were reported by the authors.

References

- Abbas, J., Aman, J., Nurunnabi, M., & Bano, S. (2019). The impact of social media on learning behavior for sustainable education: Evidence of students from selected universities in Pakistan. *Sustainability*, 11(6), 1683. <https://www.mdpi.com/2071-1050/11/6/1683>.
- Abbas, J., Balsalobre-Lorente, D., Amjid, M. A., Al-Sulaiti, K., Al-Sulaiti, I., & Aldereai, O. (2024). Financial innovation and digitalization promote business growth: The interplay of green technology innovation, product market competition and firm performance. *Innovation and Green Development*, 3(1), 100111. <https://doi.org/10.1016/j.igd.2023.100111>.
- Abbas, J., Rehman, S., Aldereai, O., Al-Sulaiti, K., I., & Syed (2023). Tourism management in financial crisis and industry 4.0 effects: Managers traits for technology adoption in reshaping, and reinventing human management systems. *Human Systems Management*, 1–18. <https://doi.org/10.3233/hsm-230067x>.
- Abbass, K., Asif, M., Niazi, A. A. K., Qazi, T. F., Basit, A., & Ahmed, A. M., F. A (2022a). Understanding the interaction among enablers of quality enhancement of higher business education in Pakistan. *PLOS ONE*, 17(5), e0267919. <https://doi.org/10.1371/journal.pone.0267919>.
- Abbass, K., Basit, A., Niazi, A. A. K., Mufti, R., Zahid, N., & Qazi, T. F. (2022b). Evaluating the social outcomes of COVID-19 pandemic: Empirical evidence from Pakistan. *Environmental Science and Pollution Research*, 30. <https://doi.org/10.1007/s11356-022-19628-7>.
- Abbass, K., Niazi, A. A. K., Qazi, T. F., Basit, A., & Song, H. (2022c). The aftermath of COVID-19 pandemic period: Barriers in implementation of social distancing at workplace. *Library Hi Tech*, 40(2), 569–585. <https://doi.org/10.1108/lht-03-2021-0113>.
- Abunamous, M. O., Boudouaia, A., Jebril, M., Diafi, S., & Zreik, M. (2022). The decay of traditional education: A case study under covid-19. *Cogent Education*, 9(1). <https://doi.org/10.1080/2331186x.2022.2082116>.
- Adiguzel, T., Kaya, M. H., & Cansu, F. K. (2023). Revolutionizing education with AI: Exploring the transformative potential of ChatGPT. *Contemporary Educational Technology*, 15(3), ep429. <https://doi.org/10.30935/cedtech/13152>.
- Ajlouni, A. O., Wahba, F. A. A., & Almahaireh, A. S. (2023). Students' attitudes towards using ChatGPT as a learning tool: The case of the University of Jordan. *International Journal of Interactive Mobile Technologies*, 17(18).
- AL-Qadri, A. H., Zhao, W., Li, M., Al-khresheh, M. H., & Boudouaia, A. (2021). The prevalence of the academic learning difficulties: An observation tool. *Heliyon*, 7, 08164. <https://doi.org/10.1016/j.heliyon.2021.e08164>.
- Al-Qadri, A. H., Zhao, W., Li, M., Al-khresheh, M. H., & Boudouaia, A. (2022). Emotional intelligence scale for international students: A proposal for a developed version. *Front Educ*, 7, 853303. <https://doi.org/10.3389/educ.2022.853303>.
- Alharbi, S., & Drew, S. (2014). Using the technology acceptance model in understanding academics' behavioural intention to use learning management systems. *International Journal of Advanced Computer Science and Applications*, 5(1), 143–155. <https://doi.org/10.14569/ijacsa.2014.050120>.
- Baidoo-Anu, D., & Ansah, L. O. (2023). Education in the era of generative artificial intelligence (AI): Understanding the potential benefits of ChatGPT in promoting teaching and learning. *Journal of AI*, 7(1), 52–62.
- Barbosa, I. (2021). Online migration of musical instrument teachers: Technology, AI, and New professional perspectives. *Communication Technologies et Développement*, 10, 1–11. <https://doi.org/10.4000/ctd.5015>.
- Bariu, T., Chun, X., & Boudouaia, A. (2022). Influence of teachers' competencies on ICT implementation in Kenyan universities. *Education Research International*, 1370052. <https://doi.org/10.1155/2022/1370052>.
- Chen, H. R., & Tseng, H. F. (2012). Factors that influence acceptance of web-based e-learning systems for the in-service education of junior high school teachers in Taiwan. *Evaluation Program Plan*, 35, 398–406.
- Cotton, D. R., Cotton, P. A., & Shipway, J. R. (2023). Chatting and cheating: Ensuring academic integrity in the era of ChatGPT. *Innovations in Education and Teaching International*, 1–12. <https://doi.org/10.1080/14703297.2023.2190148>.
- Cui, K. (2022). Artificial intelligence and creativity: Piano teaching with augmented reality applications. *Interactive Learning Environments*, 1–12. <https://doi.org/10.1080/10494820.2022.2059520>.

- Davis, F. (1986). *A technology acceptance model for empirically testing new end-user information systems: Theory and results [Doctoral Dissertation]*. Massachusetts Institute of Technology. [https://doi.org/10.1016/S0378-7206\(01\)00143-4](https://doi.org/10.1016/S0378-7206(01)00143-4)
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340. <https://doi.org/10.2307/249008>.
- Dwivedi, Y. K., Kshetri, N., Hughes, L., Slade, E. L., Jeyaraj, A., Kar, A. K., Baabdullah, A. M., Koochang, A., Raghavan, V., Ahuja, M., Albanna, H., Albashrawi, M. A., Al-Busaidi, A. S., Balakrishnan, J., Barlette, Y., Basu, S., Bose, I., Brooks, L., Buhalis, D., & Carter, L. (2023). So what if ChatGPT wrote it? Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management*, 71(0268–4012), 102642. <https://www.sciencedirect.com/science/article/pii/S0268401223000233#bib211>.
- Esling, P., & Devis, N. (2020). *Creativity in the era of artificial intelligence*. Journées d'Informatique Musicale.
- Foroughi, B., Senali, M. G., Iranmanesh, M., Khanfar, A., Ghobakhloo, M., Annamalai, N., & Bitanaghmehe-Abbaspour. (2023). Determinants of intention to use ChatGPT for educational purposes: Findings from PLS-SEM and fsQCA. *International Journal of Human–Computer Interaction*, 1–20. <https://doi.org/10.1080/10447318.2023.2226495>.
- Gilson, A., Safranek, C. W., Huang, T., Socrates, V., Chi, L., Taylor, R. A., & Chartash, D. (2023). How does ChatGPT perform on the United States medical licensing examination? The implications of large language models for medical education and knowledge assessment. *JMIR Medical Education*, 9, e45312. <https://doi.org/10.2196/45312>.
- Gomez-Ramirez, I., Valencia-Arias, A., & Duque, L. (2019). Approach to m-learning acceptance among university students: An integrated model of TPB and TAM. *International Review of Research in Open and Distributed Learning*, 20, 141–164.
- Gouzouasis, P., & Bakan, D. (2011). The future of music making and music education in a transformative digital world. *The University of Melbourne Refereed E-Journal*, 2(2), 127–154.
- Guo, K., & Wang, D. (2023). To resist it or to embrace it? Examining ChatGPT's potential to support teacher feedback in EFL writing. *Education and Information Technologies*, 1–29.
- Halaweh, M. (2023). ChatGPT in education: Strategies for responsible implementation. *Contemporary Educational Technology*, 15(2), ep421. <https://doi.org/10.30935/cedtech/13036>.
- Han, J. H., & Sa, H. J. (2021). Acceptance of and satisfaction with online educational classes through the technology acceptance model (TAM): The COVID-19 situation in Korea. *Asia Pacific Education Review*. <https://doi.org/10.1007/s12564-021-09716-7>.
- Hariri, W. (2023). Unlocking the potential of ChatGPT: A comprehensive exploration of its applications, advantages, limitations, and future directions in natural language processing. *arXiv Preprint arXiv*. <https://doi.org/10.48550/arXiv.2304.02017>. 2304.02017.
- Henriksen, D., Creely, E., Henderson, M., & Mishra, P. (2021). Creativity and technology in teaching and learning: A literature review of the uneasy space of implementation. *Educational Technology Research and Development*, 69(4), 2091–2108. <https://doi.org/10.1007/s11423-020-09912-z>.
- Ho, W. C. (2014). Music education curriculum and social change: A study of popular music in secondary schools in Beijing, China. *Music Education Research*, 16(3), 267–289.
- Ho, W. C., & Law, W. W. (2006). Challenges to globalization, localization and sinophilia in music education: A comparative study of Hong Kong, Shanghai and Taipei. *British Journal of Music Education*, 23(2), 217–237.
- Hoey, J., Schroder, T., Morgan, J., Rogers, K. B., Rishi, D., & Nagappan, M. (2018). Artificial intelligence and social simulation: Studying group dynamics on a massive scale. *Small Group Research*, 49(6), 647–683. <https://doi.org/10.1177/1046496418802362>.
- Huang, C., & Yu, K. (2021a). Research on the innovation of college music teaching mode based on artificial intelligence. *Journal of Physics: Conference Series*, 1915(2), 022051. <https://doi.org/10.1088/1742-6596/1915/2/022051>.
- Huang, Y. (2021b). Research on the application of artificial intelligence in music listening learning-taking intelligent hearing training as an example. *Journal of Physics: Conference Series*, 1915 (4).
- Haglund, J. H. (2023). *Students acceptance and use of ChatGPT in academic settings [Master thesis]*. <https://uu.diva-portal.org/smash/get/diva2:1772274/FULLTEXT01.pdf>.

- Javaid, M., Haleem, A., Singh, R. P., Khan, S., & Ibrahim Haleem Khan. (2023). Unlocking the opportunities through ChatGPT Tool towards ameliorating the education system. *BenchCouncil Transactions on Benchmarks Standards and Evaluations*, 3(2), 100115–100115. <https://doi.org/10.1016/j.bench.2023.100115>.
- Kasneci, E., Sessler, K., Küchemann, S., Bannert, M., Dementieva, D., Fischer, F., Gasser, U., Groh, G., Günnemann, S., Hüllermeier, E., Krusche, S., Kutyniok, G., Michaeli, T., Nerdel, C., Pfeffer, J., Poquet, O., Sailer, M., Schmidt, A., Seidel, T., & Stadler, M. (2023). ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and Individual Differences*, 103(102274). <https://doi.org/10.1016/j.lindif.2023.102274>.
- Khalid, A. S., Al-Sulaiti, I., & Abbas, J. (2023). Tourists' Online Information Influences their Dine-out Behaviour: Country-of-origin Effects as a Moderator. In *Country of Origin Effects on Service Evaluation* (1 ed., pp. 1–20). Qatar University Press. <https://doi.org/10.1012/qu.2023.100007>.
- Kim, G., & Lee, S. (2016). Korean students' intentions to use mobile-assisted language learning: Applying the technology acceptance model. *International Journal of Contents*, 12(3), 47–53. <https://doi.org/10.5392/ijoc.2016.12.3.047>.
- Kirlidog, M., & Kaynak, A. (2013). Technology acceptance model and determinants of technology rejection. *International Journal of Information Systems and Social Change*, 2(4), 1–12. <https://doi.org/10.4018/jissc.2011100101>.
- Klanjscek, N., David, L., & Frank, M. (2021). Evaluation of an e-learning tool for augmented acoustics in music education. *Music & Science*, 4, 205920432110375–205920432110378. <https://doi.org/10.1177/20592043211037511>.
- Kohnke, L., Moorhouse, B. L., & Zou, D. (2023). Exploring generative artificial intelligence preparedness among university language instructors: A case study. *Computers and Education: Artificial Intelligence*, 5, 100156.
- Lund, B. D., & Wang, T. (2023). Chatting about ChatGPT: How may AI and GPT impact academia and libraries? *Library Hi Tech News*. <https://doi.org/10.1108/LHTN-01-2023-0009>.
- Lee, B. C., Yoon, J. O., & Lee, I. (2009). Learners' acceptance of e-learning in South Korea: Theories and results. *Computers & Education*, 53(4), 1320–1329. <https://doi.org/10.1016/j.compedu.2009.06.014>.
- Lee, C., Yeung, A. S., & Ip, T. (2017). University English language learners' readiness to use computer technology for self-directed learning. *System*, 67, 99–110. <https://doi.org/10.1016/j.system.2017.05.001>.
- Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 46(1), 50–80. <https://doi.org/10.1518/hfes.46.1.50.30392x>.
- Lee, Y. C. (2020). Artificial intelligence and continuous usage intention: Evidence from a Korean online job information platform. *Business Communication Research and Practice*, 3(2), 86–95. <https://doi.org/10.22682/bcrp.2020.3.2.86>.
- Li, N., & Ismail, M. J. B. (2022). Application of artificial intelligence technology in the teaching of complex situations of folk music under the vision of new media art. *Wireless Communications and Mobile Computing*, 1–10. <https://doi.org/10.1155/2022/5816067>.
- Li, P., & Wang, B. (2023). Artificial Intelligence in music education. *International Journal of Human-Computer Interaction*, 1–10. <https://doi.org/10.1080/10447318.2023.2209984>.
- Liu, G., & Ma, C. (2023). Measuring EFL learners' use of ChatGPT in informal digital learning of English based on the technology acceptance model. *Innovation in Language Learning and Teaching*, 1–14. <https://doi.org/10.1080/17501229.2023.2240316x>.
- Lupker, J. A. T., & Turkel, W. J. (2021). Music theory, the missing link between music-related big data and artificial intelligence. *Digital Humanities Quarterly*, 15(1), 1–2.
- Maqsood, A., Abbas, J., Rehman, G., & Mubeen, R. (2021). The paradigm shift for educational system continuance in the advent of COVID-19 pandemic: Mental health challenges and reflections. *Current Research in Behavioral Sciences*, 2, 100011. <https://doi.org/10.1016/j.crbeha.2020.100011>.
- McGee, R. W. (2023). Using Artificial Intelligence (AI) to compose a musical score for a Tai Chi tournament routine: A ChatGPT experiment. Working Paper. <https://doi.org/10.13140/RG.2.2.24656.99843>.
- Meng, Q., Yan, Z., Abbas, J., Shankar, A., & Subramanian, M. (2023). Human-computer interaction and digital literacy promote educational learning in pre-school children: Mediating role of psychological resilience for kids' mental well-being and school readiness. *International Journal of Human-Computer Interaction*, 1–15. <https://doi.org/10.1080/10447318.2023.2248432x>.
- Mijwil, M., Aljanabi, M., & Ali, A. H. (2023). ChatGPT: Exploring the role of cybersecurity in the protection of medical information. *Mesopotamian Journal of Cyber Security*, 18–21. <https://doi.org/10.58496/MJCS/2023/004>.

- Mizumoto, A., & Eguchi, M. (2023). Exploring the potential of using an AI language model for automated essay scoring. *Research Methods in Applied Linguistics*, 2(2), 100050.
- NeJhaddadgar, N., Ziapour, A., Zakkipour, G., Abbas, J., Abolfathi, M., & Shabani, M. (2020). Effectiveness of telephone-based screening and triage during COVID-19 outbreak in the promoted primary healthcare system: A case study in Ardabil province, Iran. *Journal of Public Health* <https://doi.org/10.1007/s10389-020-01407-8x>.
- Oforu-Ampong, K., Acheampong, B., Kevor, M. O., & Amankwah-Sarfo, F. (2023). Acceptance of artificial intelligence (ChatGPT) in education: Trust, innovativeness and psychological need of students. *Information and Knowledge Management*, 13(4). <https://doi.org/10.7176/ikm/13-4-03>.
- Peng, R., Hu, Q., & Kouider, B. (2023a). Teachers' acceptance of online teaching and emotional labor in the EFL context. *Sustainability*, 15(18), 13893–13893. <https://doi.org/10.3390/su151813893x>.
- Perkins, M. (2023). Academic integrity considerations of AI large language models in the post-pandemic era: ChatGPT and beyond. *Journal of University Teaching and Learning Practice*, 20(2). <https://doi.org/10.53761/1.20.02.07x>.
- Quan, Y. (2019). Applied research of artificial intelligence in music education. *Northern Music*, 17, 38–44. <https://doi.org/10.2991/assehr.k.200401.012>.
- Rahmat, T. E., Raza, S., Zahid, H., Abbas, J., Mohd Sobri, F. A., & Sidiki, S. N. (2022). Nexus between integrating technology readiness 2.0 index and students' e-library services adoption amid the COVID-19 challenges: Implications based on the theory of planned behavior. *Journal of Education and Health Promotion*, 11, 50. https://doi.org/10.4103/jehp.jehp_508_21.
- Rudolph, J., Tan, S., & Tan, S. (2023). ChatGPT: Bullshit spewer or the end of traditional assessments in higher education? *Journal of Applied Learning & Teaching*, 6(1), 1–22. <https://doi.org/10.37074/jalt.2023.6.1.9>.
- Russell, S., & Norvig, P. (2020). *Artificial intelligence: A modern approach (4th ed.)*. Pearson.
- Sánchez, R. A., & Hueros, A. D. (2010). Motivational factors that influence the acceptance of Moodle using TAM. *Computers in Human Behavior*, 26(6), 1632–1640. <https://doi.org/10.1016/j.chb.2010.06.011>.
- Schmidt-Fajlik, R. (2023). ChatGPT as a grammar checker for Japanese english language learners: A comparison with grammarly and proWritingAid. *AsiaCALL Online Journal*, 14(1), 105–119. <https://doi.org/10.54855/acoj.231417>.
- Shah, S. N., Fahlevi, M., Rahman, E. Z., Akram, M. S., Jamshed, K., Aljuaid, M., & Abbas, J. (2023). Impact of green servant leadership in Pakistani small and medium enterprises: Bridging pro-environmental behaviour through environmental passion and climate for green creativity. *Sustainability*, 15(20), 14747–14747. <https://doi.org/10.3390/su152014747>.
- Strzelecki, A. (2023). To use or not to use ChatGPT in higher education? A study of students' acceptance and use of technology. *Interactive Learning Environments*, 1–14. <https://doi.org/10.1080/10494820.2023.2209881x>.
- Tsai, B. Y., & Yen, J. N. (2014). The influence of self-efficacies on readers' intention to use e-Reading devices: An empirical study. *International Journal of Distance Education Technologies*, 12(3), 41.
- Venkatesh, V., & Davis, F. D. (2000). A theoretical extension of the technology acceptance model: Four longitudinal field studies. *Management Science*, 46(2), 186–204.
- Venkatesh, V., Thong, J. Y. L., & Xu, X. (2012). Consumer acceptance and use of information technology: Extending the unified theory of acceptance and use of technology. *MIS Quarterly*, 36(1), 157–178.
- Warnerfjord, M. (2023). *Evaluating ChatGPT's ability to compose music using the MIDI file format* degree project in computer science thesis. Kth Royal Institute of Technology.
- Wei, J., Marimuthu, K., & Prathik, A. (2022). College music education and teaching based on AI techniques. *Computer and Electrical Engineering*, 100, 107851.
- Xiao, Y., & Zhi, Y. (2023). An exploratory study of EFL learners' use of ChatGPT for language learning tasks: Experience and perceptions. *Languages*, 8(3), 212. <https://doi.org/10.3390/languages8030212>.
- Xie, J., & Leung, B. W. (2011). Students' motivation to study music: The mainland China context. *Research Studies in Music Education*, 33(1), 59–72.
- Xie, Y., Boudouaia, A., Xu, J., AL-Qadri, A. H., Khattala, A., Li, Y., & Aung, Y. M. (2023). A study on teachers' continuance intention to use technology in English instruction in western China Junior secondary schools. *Sustainability*, 15(5), 4307. <https://doi.org/10.3390/su15054307>.
- Xu, Q., & Boudouaia, A. (2023). A study on technology use for sustainable graduate education internationalization at home: Chinese teachers' experiences and perspectives. *Sustainability*, 15(13), 10621. <https://doi.org/10.3390/su151310621x>.

- Yao, J., Ziapour, A., Abbas, J., Toraji, R., & NeJhaddadgar, N. (2022). Assessing puberty-related health needs among 10–15-year-old boys: A cross-sectional study approach. *Archives de Pédiatrie*<https://doi.org/10.1016/j.arcped.2021.11.018x>.
- Ye, F. (2020). A study on music education based on artificial intelligence. *IOP Conference Series: Materials Science and Engineering*, 750(1), 012115. <https://doi.org/10.1088/1757-899X/750/1/012115/x>.
- Zhai, X. (2022). ChatGPT user experience: Implications for education. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.4312418>
- Zhang, J., & Wan, J. (2020a). A summary of the application of artificial intelligence in music education. *International Conference on education, economics and information management (ICEEIM 2019)* (pp. 42–44). Atlantis Press.
- Zhang, W. (2020b). A study on the user acceptance model of artificial intelligence music based on UTAUT. *Journal of the Korea Society of Computer and Information*, 25(6), 25–33. <https://doi.org/10.9708/jksoci.2020.25.06.025>.
- Zhang, Z. (2023). The application and research of artificial intelligence in the field of music education. *Proceedings of the 2022 3rd International Conference on Artificial Intelligence and Education (IC-ICAIE 2022)*, 297–302. https://doi.org/10.2991/978-94-6463-040-4_45.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Wang Zhou¹  · YeaJin Kim¹

✉ YeaJin Kim
ykim114@gmail.com

Wang Zhou
Zhouwangssci@126.com

¹ Chugye University for the Arts, Seoul, Korea